AI Agents Illustrated by a Robot Vacuum

Introduction

Artificial Intelligence (AI) agents are systems that perceive their environment, reason about it, and act to achieve goals. A robot vacuum provides a concrete, everyday example of such an agent, showing how perception, decision■making, and action combine to produce intelligent behavior.

What Is an AI Agent?

An AI agent consists of sensors (to gather data), an internal state or model (its belief about the world), a goal or utility function (what it wants to achieve), a policy (how it decides what to do), and actuators (the means to affect the environment). The agent continuously cycles through perception, reasoning, and action to fulfill its objectives.

Robot Vacuum as an AI Agent

Sensors: LiDAR, cameras, infrared, bumpers, cliff detectors, and dirt sensors provide raw data about obstacles, floor layout, and cleanliness. Internal State: a map of the home, battery level, dust■bin status, and a record of cleaned areas. Goal: maximize floor coverage while avoiding obstacles and preserving battery life. Policy: path■planning algorithms (e.g., SLAM, coverage planning) and reactive rules (e.g., turn away on collision). Actuators: wheel motors, suction motor, brushes, docking mechanism.

How Intelligence Manifests

Autonomy: the vacuum starts, cleans, and returns to its dock without human input. Planning: it builds a map and computes an efficient coverage path. Adaptation: it learns dirt■hotspot locations and adjusts future cleaning patterns. Multi■objective optimization: balances cleaning thoroughness, battery consumption, and time.

Example Walkthrough

1. Startup – sensors scan the room and create a map. 2. Goal selection – "clean all reachable floor area." 3. Planning – compute a zig■zag coverage path. 4. Execution – move, vacuum, and react to obstacles. 5. Learning – record high■dirt zones for extra passes later. 6. Battery management – when battery <20%, navigate to dock, recharge, and resume remaining area.

Conclusion

The robot vacuum encapsulates the core components of an AI agent: perception, internal modeling, goal■directed decision making, and action. By observing how it autonomously maps, plans, reacts, and learns, we gain a clear, tangible understanding of AI agents in practice.